

INVENTORY STOCK ANALYSIS SYSTEM

Python Programming Lab (N-PCCCD304P) • Experiential Learning Utility

Utility Assessment – I

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The Problem

Small retail businesses still track inventory manually using registers or spreadsheets — an approach that invites errors, hides low-stock items until it's too late, and leaves sales data scattered and hard to analyze.

The Inventory Stock Analysis System automates this — tracking stock, logging sales, raising restock alerts, and generating analysis reports.



1

Manual tracking

Registers & spreadsheets

2

Errors & delays

Low stock goes unnoticed

3

Lost sales

Business impact

Objectives



1 Store & organize item-wise stock, price and details

2 Update stock in real time — purchases in, sales out

3 Auto-generate restock alerts below threshold

4 Analyze stock using NumPy & Pandas

5 Persist data permanently via file handling

Proposed Solution



A menu-driven Python application built around modular, reusable functions. Each inventory item is one record in a list; core operations — add, update, sell, alert — are separate functions kept small and testable.

Scope

Single-user, command-line utility with local file storage (CSV/JSON).
Multi-user access and a GUI are future scope.

Built with

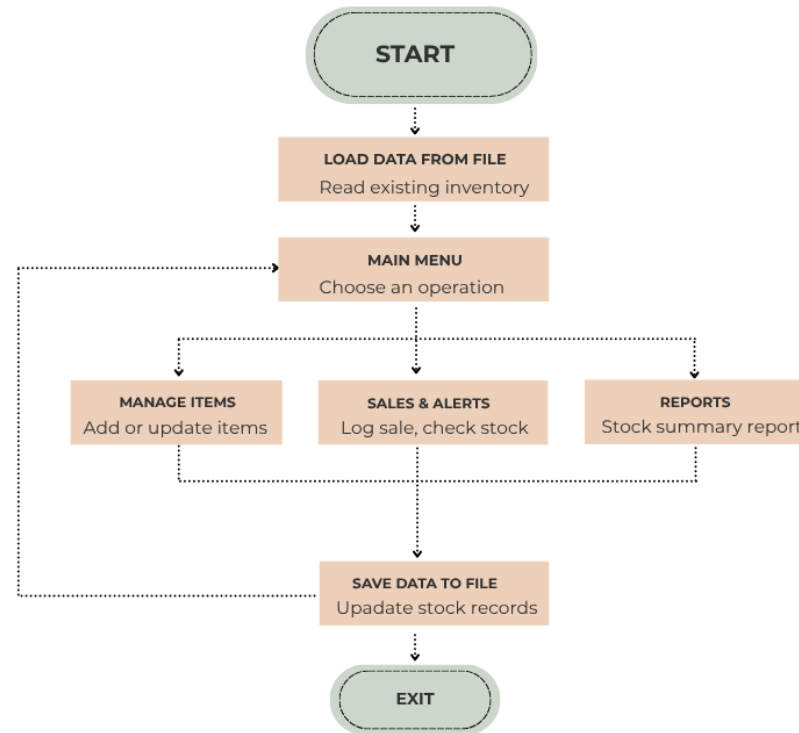
● **Lists & Dictionaries**

● **Modular Functions**

● **NumPy + Pandas**

● **File Handling (JSON)**

Program Flow



Module-wise Breakdown



How the five Python Lab modules map onto this utility

Module 1	Fundamentals / Control Structure	Menu loop, if-elif choice handling
Module 2	Data Structures	Inventory as a list of dictionaries
Module 3	Functions	One function per operation (add, update, sell...)
Module 4	NumPy, Pandas	Stock statistics & summary report
Module 5	File Handling & OOP	JSON-based permanent storage

Functions Implemented



`load_data()`

Reads inventory from file at startup

`add_item()`

Adds a new item to inventory

`record_sale()`

Logs a sale transaction

`generate_summary()`

NumPy/Pandas stock summary

`save_data()`

Writes inventory back to file

`update_stock()`

Updates qty on purchase / sale

`check_restock_alert()`

Flags items below threshold

`main()`

Menu loop & routing

Version Control Workflow



Repo Setup

Public GitHub repo + README

Skeleton Commit

Menu structure & empty functions

Docs Added

README + flowchart pushed

Tracked Progress

Descriptive commits, main branch stable

github.com/anshulkumbhare18/inventory-stock-analysis-system

Public repository • README.md • Feature branches planned for Phase 2

What's Next

Phase 2 (Week 2)

Implement all functions, use feature branches, commit progress regularly

Phase 3 (Week 3)

Finish NumPy/Pandas analysis, test thoroughly, finalize report & demo

Thank You

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